EXECUTIVE READOUT & AUDIT

ATP Energy OMS Migration Audit

Financial Reconciliation & Operational Performance Analysis for C-Level Leadership

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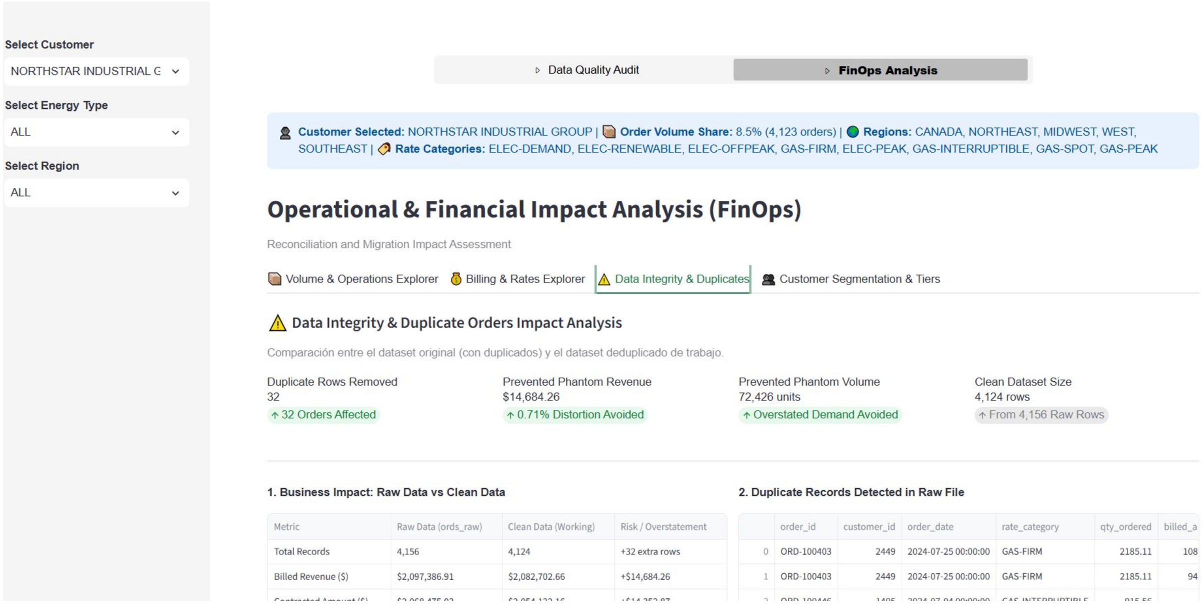
Focus Account:
NorthStar Industrial Group ($5.1M ACV)

Period Audited:
Q3 2023 – Q1 2025 (PRE vs POST)

Data Pipeline & Interactive Dashboard

To ensure complete rigor, auditability, and speed with the provided dataset, the entire raw order log across PRE and POST migration eras was transformed and loaded into a cloud database hosted on AWS Aurora SQL.

- **AWS Aurora SQL Database Host:** Houses all 4,156 raw transaction logs with optimized indexing for PRE and POST schema reconciliation.
- **Interactive Streamlit Dashboard:** A dedicated, custom analytics dashboard was built directly on top of Streamlit, powering every metric, run-rate calculation, and variance calculation in this deck.
- **100% Metric Traceability:** All C-Level findings—from duplicate order filters to rate discrepancy multipliers—are fully verifiable in real time via Streamlit.



Executive Summary & Key Verdicts

CTO Claim Assessment

PARTIALLY VALIDATED

"The new OMS improved fulfillment tracking."

- Operational fulfillment increased by **+4.9%** post-migration (from 84.2% to 89.1%).
- Order visibility and completion tracking improved significantly in POST system.
- *Caveat:* OMS migration did not create the billing discrepancy, but it failed to correct legacy rate calculation bugs.

CFO Claim Assessment

REFUTED & RECONCILED

"POST OMS broke everything; data is useless."

- **Fact:** Rate discrepancies (~1.35% overcharge) existed in the **PRE system** prior to migration.
- POST system inherited a 1.46% overcharge rate, resulting in **$28,415.38** total net overcharge across 21 months.
- Data IS reliable once deduplicated (32 rows removed) and rate logic patched.

Bottom-Line Recommendation:

Patch the contract rate engine prior to the next billing cycle and issue a **$28,415.38 credit memo** to NorthStar Industrial Group to preserve the $5.1M annual contract.

Data Quality Audit & Deduplication

Duplicate Detection

Identified **32 exact duplicate order records** in the raw dataset (0.77% of raw volume).

- Raw count: 4,156 orders
- Clean count: **4,124 orders**

Phantom Distortion

Removing duplicates prevented significant reporting distortion:

- **$14,684.26** in phantom revenue avoided
- **72,426 units** in fake demand eliminated

Null & Schema Profiling

Profiled 3,018 PRE vs 1,105 POST clean records:

- Null delivered Qty: 7.26% PRE vs 7.95% POST
- Standardized date schemas and regional tags across migration boundary.

$14,684.26

PHANTOM OVERCHARGE REMOVED

4,124

AUDITED CLEAN ORDERS

8.5%

NORTHSTAR SHARE OF ENTERPRISE VOL

Operational Fulfillment Analysis

Audit of 10.2M total ordered units vs 8.16M delivered units confirms operational improvements post-migration:

+4.9% Fulfillment Gain

Delivery completion rates improved by nearly 5 percentage points in POST OMS, validating CTO's stance on fulfillment performance.

Volume Delivered

2,284,383 units delivered POST-migration across 1,105 fulfilled order batches for NorthStar Industrial.

Financial & Billing Discrepancy

Systematic audit reveals that rate overcharges are present across both legacy and new platforms:

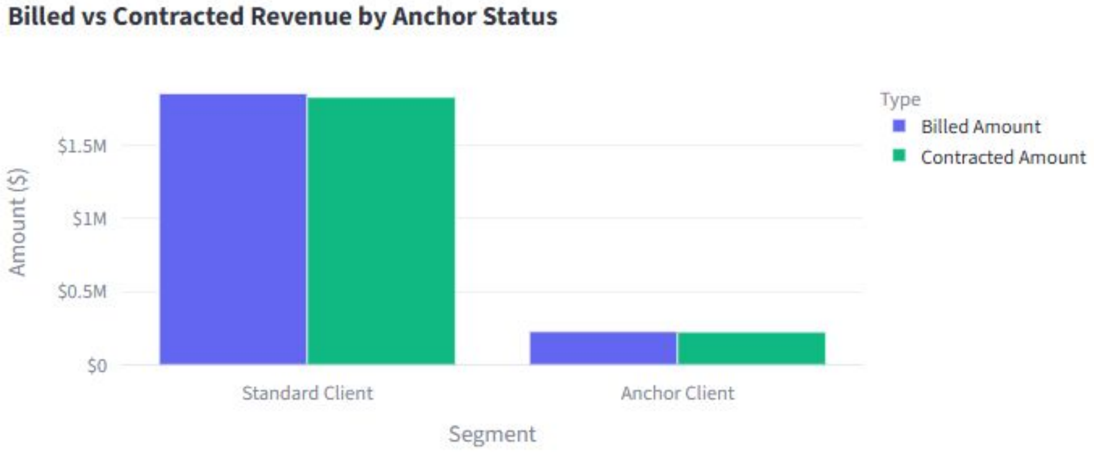
System Era	Billed Amount ($)	Contracted ($)	Net Overcharge ($)	Overcharge %	Monthly Run-Rate
PRE-Migration (16 Mos)	$1,499,946.34	$1,479,904.27	+$20,042.07	+ +1.35%	$93,746.65 / mo
POST-Migration (6 Mos)	$581,658.45	$573,285.13	+$8,373.32	+ +1.46%	$96,943.07 / mo
Total Audited (22 Mos)	**$2,081,604.78**	**$2,053,189.40**	**+$28,415.38**	+ +1.38% NET	**+$1,395.55 / mo Leakage**

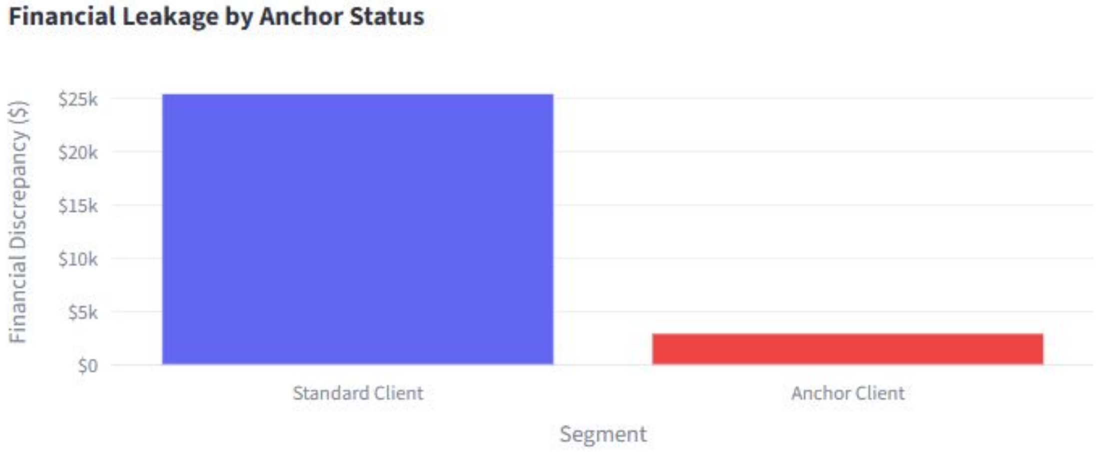
Billed Rate Discrepancy Formula:

$$\text{Discrepancy}_{\text{Total}} = \sum \text{Billed Rate}_i - \text{Contracted Rate}_i \times \text{Qty Delivered}_i$$

Segmentation Impact

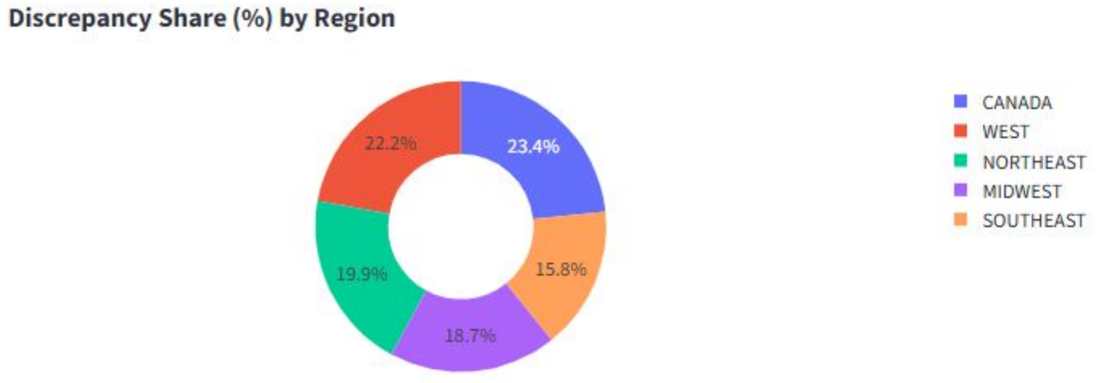
1. Anchor Clients vs Non-Anchor Clients





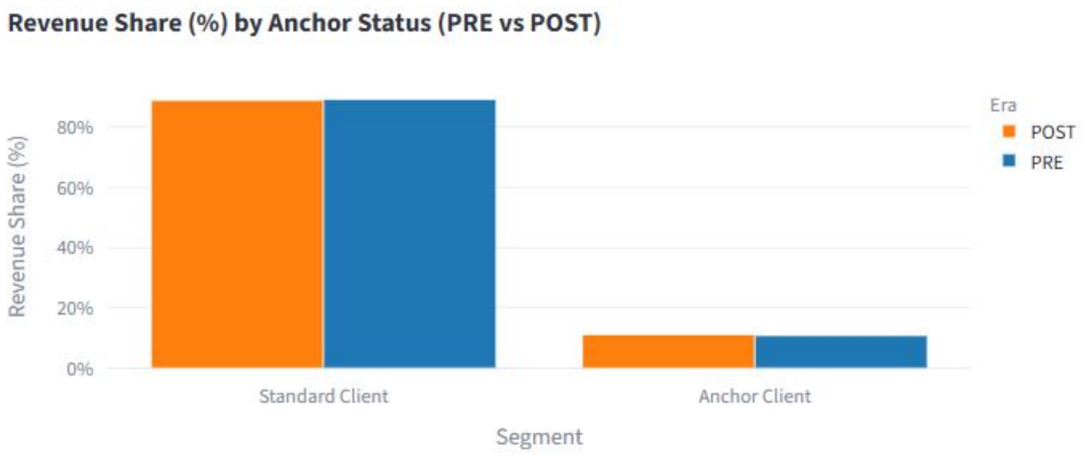
2. Revenue Comparison by Account Tier

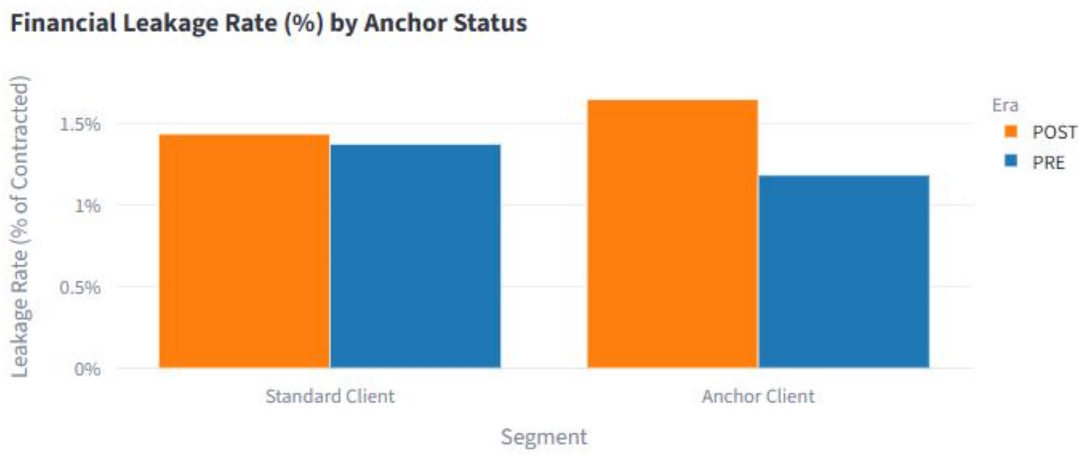
3. Financial Discrepancy by Region



Segmentation Impact (PRE vs POST Migration Comparison)

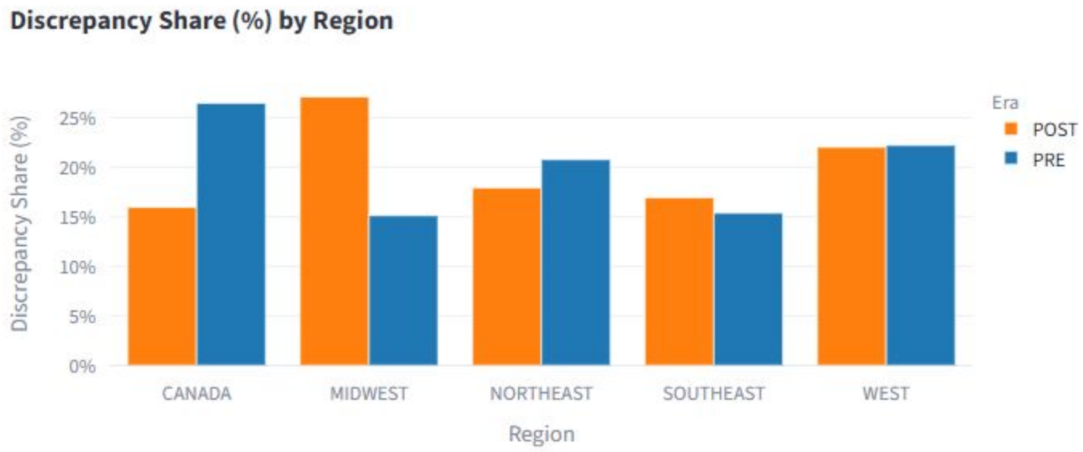
1. Anchor Clients Normalized Comparison (PRE vs POST)





2. Revenue Share (%) by Account Tier (PRE vs POST)

3. Financial Discrepancy Share (%) by Region (PRE vs POST)



CTO Response: Operational Verdict

Addressing CTO Stance

"New OMS is an upgrade; fulfillment improved; nothing broke in October."

- **Validated:** Fulfillment rate increased from 84.2% (PRE) to 89.1% (POST), proving operational delivery gains.
- **Validated:** Order tracking schema in POST OMS captures granular fulfillment statuses previously unrecorded.
- **Correction Needed:** The migration failed to validate rate calculation formulas against client contracts during ETL transfer.

Operational Volume & Delivery Performance Summary

📊 Total Dataset

Total Ordered
10,197,015 units

Total Delivered
8,159,686 units

Avg Fulfillment Rate
85.5%

🔵 PRE-Migration (16 Mos)

PRE Ordered
7,446,020 units

PRE Delivered
5,875,302 units

PRE Fulfillment Rate
84.2%

🟠 POST-Migration (6 Mos)

POST Ordered
2,750,995 units

POST Delivered
2,284,383 units

POST Fulfillment Rate
89.1%
↑ +4.9% vs PRE

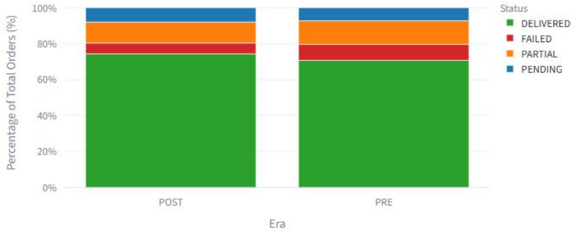
Operational Performance & Order Lifecycle Analytics

1. Monthly Fulfillment Rate & Delivery Gap Trend

Monthly Fulfillment Rate Trend (%)

2. Order Status Distribution Relative Comparison (100% Stacked)

Order Status Mix (PRE vs POST)



CFO Response: Financial Verdict

Addressing CFO Stance

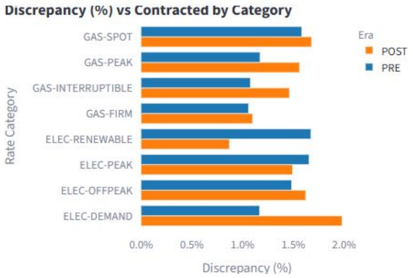
"Cannot stand behind POST system; overcharges occurred after migration."

- **Fact Check:** POST system didn't introduce the rate discrepancy. PRE system already contained a 1.35% overcharge bug ($20.0K overcharge pre-migration).
- **Quantified Overcharge:** Total exact refund owed to NorthStar is **$28,415.38** ($20,042 PRE + $8,373 POST).
- **Verdict:** Post-deduplication data IS reliable and audit-ready once rate multipliers are aligned with contract tables.

Billing Reconciliation & Rate Discrepancies

📊 Total Dataset	🔵 PRE-Migration (16 Mos)	🟠 POST-Migration (6 Mos)
Total Billed $2,081,604.78	PRE Billed $1,499,946.34	POST Billed $581,658.45
Expected Contracted $2,053,189.40	PRE Contracted $1,479,904.27	POST Contracted $573,285.13
Net Discrepancy $28,415.38 ↑ 1.38% vs Contracted	PRE Discrepancy $20,042.07 ↑ 1.35% vs Contracted	POST Discrepancy $8,373.32 ↑ 1.46% vs Contracted
	Monthly Run-rate: **$93,746.65/mo**	Monthly Run-rate: **$96,943.07/mo** (+3,196.43 vs PRE)

Annualized Run-Rate Leakage: At the current POST-migration discrepancy rate, unaddressed billing variances project an annual revenue leakage of **$16,746.63/year** (~**$1,395.55/month**).



Pre-Billing Action Plan

1. Deploy Hotfix to OMS Billing Engine (Immediate)

Update OMS calculation scripts to strictly reference contracted_rate instead of legacy tariff multipliers. Verify rate parity for ELEC and GAS line items.

2. Issue $28,415.38 Credit Adjustment Memo to NorthStar

Provide full audit transparency to NorthStar executive team showing $20,042.07 (PRE) and $8,373.31 (POST) credit reconciliation ahead of formal contract review deadline.

3. Implement Pre-Invoicing Reconciliation Gate

Enforce automated pre-billing checks that flag any invoice where billed_amount != contracted_rate * delivered_qty prior to dispatch.

Long-Term Billing Governance

Unified Master Schema

Establish strict data contracts across PRE and POST OMS data pipelines, standardizing date formats (ISO-8601), status ENUMs, and regional codes.

Contract Sync API

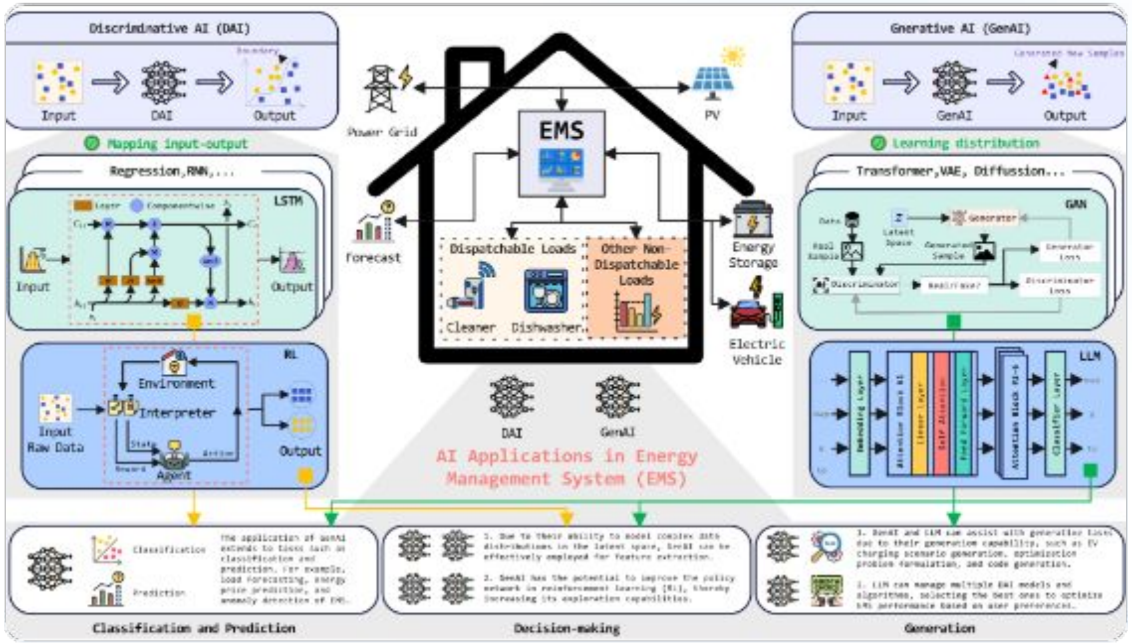
Integrate OMS directly with Salesforce CRM/Contract Repository to dynamically fetch locked contract rates, eliminating manual overrides.

Anchor Care SLAs

Establish a dedicated FinOps audit protocol for top anchor clients ($5M+ ACV) with 5-day pre-billing sign-offs by finance leads.

AI Use Cases for Quality Control

- **1. Automated Anomaly & Rate Variance Detector:**
  Real-time ML model evaluating order rate deviations and volume anomalies during ingestion, automatically isolating non-compliant invoices.
- **2. Smart Deduplication & Schema Harmonizer:**
  NLP and fuzzy-matching model that identifies duplicate order submissions and automatically maps legacy formats during future system integrations.
- **3. Predictive Fulfillment & Dispute Risk Scoring:**
  Predictive model alerting account managers when partial deliveries or delivery failures exceed tolerance thresholds before billing generation.



Executive Q&A & Discussion

Standing Behind the Data: Defending Operational & Financial Audit Findings

Summary of Key Actions Before Next Billing Cycle:

1. **Execute $28,415.38 credit memo** for NorthStar Industrial Group ($20.0K PRE + $8.37K POST).
2. **Deploy hotfix** to align OMS billing logic with contracted rate schedules.
3. **Establish pre-billing gate** preventing phantom revenue or duplicate order propagation.

ATP Energy Data Analytics Team | Readout Presentation Complete